Adverk Technologies

IOT & ROBOTICS

Project Name : Home Automation System using HC-05 Bluetooth Module.

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Abstract

This project details the creation and deployment of a home automation system that leverages the HC-05 Bluetooth module. The system enables users to manage household electrical appliances through a smartphone application, offering an economical and readily available smart home solution. It incorporates Arduino as the microcontroller, relays for switching, and an Android-based user interface, resulting in an affordable wireless automation system. This approach capitalizes on the increasing popularity of smart homes, driven by their convenience, security, and energy efficiency.

Objective

Develop a cost-effective and user-friendly wireless home automation system. This system will utilize Bluetooth technology to enable remote control of essential home appliances (e.g., lights, fans) via an Android smartphone. The primary goals are to enhance comfort and accessibility, particularly for elderly and physically challenged individuals, through an easy-to-install hardware setup.

Problem Statement

Many households continue to operate appliances manually, despite advancements in technology. This results in several challenges:

- **Energy Inefficiency:** Forgetting to turn off devices leads to wasted energy.
- **Inconvenience:** Physically challenged individuals and the elderly face difficulties with manual control.
- **Costly Smart Home Integration:** Affordable smart home solutions are often unavailable.
- **System Limitations:** Conventional systems are hindered by complex wiring and limited scalability.

Methodology

Hardware Components

- **Arduino Uno:** Microcontroller for processing commands.
- **HC-05 Bluetooth Module:** Enables wireless communication.
- **Relay Module:** Controls high-voltage appliances.
- **Smartphone (Android):** Serves as the user interface via an application.
- **Power Supply & Wiring:** Provides safe power to the system.

Methodology

Software

- Arduino IDE: Used for writing and uploading code to the microcontroller.
- MIT App Inventor / Android Studio: Used for developing the Android application.
- Bluetooth Serial Communication: Used for transmitting data from the phone to the Arduino.

Methodology

3. Working Principle

1. The user initiates the process by opening the mobile application and pairing it with the HC-05 module via Bluetooth.
2. Upon pressing a button within the app (e.g., "Turn on Light"), a specific character (e.g., '1') is transmitted.
3. The Arduino receives this character and subsequently activates the corresponding relay.
4. Consequently, the appliance is switched ON or OFF as directed.

Result

This system successfully enabled wireless control of multiple home appliances via Bluetooth. Key performance indicators include:

- Responsiveness: Minimal delay (~1 second) between command and execution.
- Range: Effective operation within a 10-meter Bluetooth range.
- Usability: An intuitive and user-friendly interface.
- Cost-Effectiveness: Low implementation costs.

Advantages

- Low cost and simple implementation.
- No need for internet connection.
- Easy-to-use interface for all age groups.
- Customizable and scalable.

Limitation

- Limited Bluetooth range (approximately 10 meters).
- Requires proximity and the app to be open to function.
- Lacks a feedback mechanism, making command execution unconfirmable without visual verification.

Future Scope

To achieve long-range operation, incorporate Wi-Fi control using NodeMCU or ESP8266. Integrate voice control capabilities through Google Assistant or Alexa. Implement real-time feedback mechanisms by utilizing sensors and IoT dashboards. Enhance security by adding authentication features within the application.

Conclusion

This project showcases an accessible and affordable home automation system. Utilizing the HC-05 Bluetooth module, it effectively integrates essential household functions with contemporary automation, making smart home technology readily available to a broader audience.